

Data best practices

Why best practices?

- Be able to explain what you did
- Lets someone else (or you) reproduce/bulletproof your work
- For longer projects, you might not actually remember all of your steps
- Your process might be useful to you (or someone else) again someday

Step 1: Keep it clean

Download and save an original copy of your data – direct from the source, with the name as it is, and no alterations.

Whenever you make changes, make and save a COPY of the file. I like to add a suffix to the filename that describes the changes. (-deduped.csv, -sorted.csv, -cleaned.csv)

If my project involves a lot of changes to the data, I'll add a date to the suffix.

Step 2: Read the data dictionary

Read the documentation that goes with the data file.

This could be a data dictionary:

<https://www.cms.gov/Research-Statistics-Data-and-Systems/Files-for-Order/LimitedDataSets/Downloads/InpatientVersionJ2011.pdf>

Or a website:

http://irsa.ipac.caltech.edu/applications/DDGEN/Doc/dd_tbl.html#examples

... Or something else entirely. (For example, maybe it's in an email to a government worker.) But you will want to know what each field means and how it's coded. Keep track of this.

[Here's the Open Data DC documentation for the 311 service requests](#)

Step 3: Create a data diary

Create a text document, if you're working in Sheets or Excel, a Google doc works well. Or Sublime Text!

Name it something useful and put it where you can find it.

Put your name, the date and a short description of what the project is at the top.

If you don't have a complete analysis plan from the outset, that's okay. You can circle back to this description at the end and make it more accurate and descriptive.

Step 4: Count!

Count the number of rows/records in your data. Make sure this matches the number (hopefully) described in the documentation. If so, then enter the record count in your text document.

311 Example:

Record count: The 311 sample dataset contains 9,153 records. You can look at row numbers, but exclude the header row. Or you can highlight a column that has no blank records (but again subtract 1 for the header row).

Step 5: Keep track of changes

Track your work.

Example:

The first thing I wanted to know was when the most recent request came in. So, I clicked on column M (ADDDATE) to select it.

Data -> Sort sheet by column M, Z to A.

That goes in your data diary.

Step 6: Show your output

Where applicable include the output as well. (It's often OK to truncate).

Example:

These are the top rows after the sort.

H	I	J	K	L	M	N	O
SERVICECODE	SERVICECODEDESCRIPTION	SERVICETYPECODEDESCRIPTION	ORGANIZATIONACRON	SERVICECA...	ADDDATE	RESOLUTIONDATE	SERVICEDUEDATE
S0477	Scheduled Yard Waste	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:23:...	2026/05/06 16:40:...	2026/05/07 03:59:...
RPP	Residential Parking Permit Vio...	PEMA- Parking Enforcement M.	DPW	1	2026/04/01 11:22:...	2026/04/01 14:27:...	2026/04/02 11:22:...
S0000	Abandoned Vehicle - On Public ...	PEMA- Parking Enforcement M.	DPW	1	2026/04/01 11:21:...	2026/05/18 13:35:...	2026/04/21 14:55:...
S0441	Trash Collection - Missed	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:20:...	2026/04/02 14:42:...	2026/04/03 11:20:...
S0031	Bulk Collection	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:20:...	2026/04/08 15:37:...	2026/04/09 03:59:...
S0477	Scheduled Yard Waste	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:20:...	2026/05/08 16:11:...	2026/05/09 03:59:...
S0441	Trash Collection - Missed	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:19:...	2026/04/02 18:07:...	2026/04/03 11:19:...
S0321	Recycling Collection - Missed	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:16:...	2026/04/01 12:10:...	2026/04/03 11:16:...
S0459	Tree Removal	Urban Forrestry	DDOT	1	2026/04/01 11:15:...	2026/04/01 12:51:...	2026/12/21 11:15:...
S0311	Rodent Inspection and Treatment	DOH- Department Of Health	DOH	1	2026/04/01 11:14:...	2026/04/01 13:49:...	2026/04/08 11:14:...
S0031	Bulk Collection	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:13:...	2026/04/08 12:35:...	2026/04/09 03:59:...
S0031	Bulk Collection	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:12:...	2026/04/06 15:37:...	2026/04/07 03:59:...
S0181	Illegal Dumping	SWMA- Solid Waste Manageme	DPW	1	2026/04/01 11:12:...	2026/04/01 13:54:...	2026/04/21 11:12:...

When including results is impractical, note the filename and spreadsheet URL. It can also be useful to include row counts. The goal is for someone else to make sure they can come up with the same answer as you.

More Step 6

If you used a function, copy and paste the exact text of the function you used.

Example:

Use this function to count how many requests came from Ward 7:

```
=countif(X:X, "Ward 7")
```

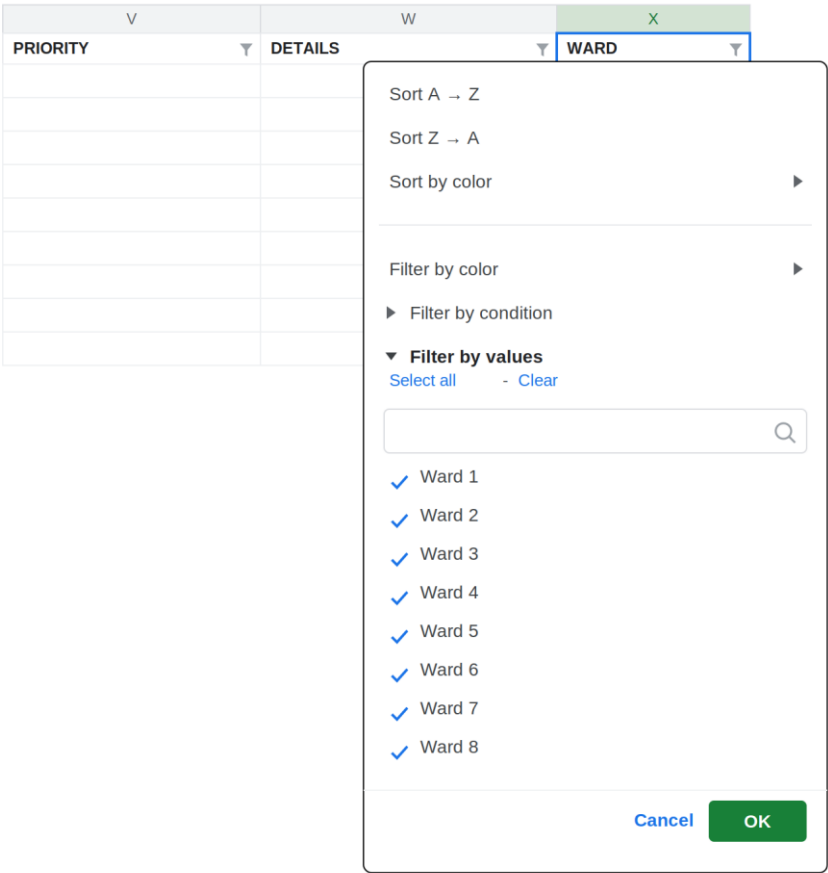
If can also be useful to note characteristics of fields.

(Column M, "ADDDATE," ranges from 2026/03/25 to 2026/04/01.)

More detail

If you are using dropdown menus, it can also be useful to take a screenshot.

But screenshots by themselves aren't enough; have a description, too.



Review: Why we are bothering with this

- You will know what you did
- You can easily do it again
- Someone can check your work
- If necessary, you can share what you did with experts
- It will make it easy to share what you did with the subject of a story
- It will make it easy to share what you did with readers
- If your work comes under question, you can show that you did due diligence